

Starting a new look for these notes. I don't want it to feel systematic. Rather, personal. Sorry if this interrupts anyone who has been learning with me. - Kevin

Active Directory Objects, Identity, and Enterprise Administration

Goal: Understand: What is AD actually storing and managing?

Yesterday: Domain Controller, Today: The contents of the Domain Controller

Active Directory Objects

An Active Directory object is any item that Active Directory stores information about and manages. Think AD = giant database for a Windows network. Every thing inside that database is an object:

User - employee accounts (e.g., Kevin)

The objects are a good, structured way to represent and manage everything on a network. It stores users, computers, printers, and groups as objects with its own attributes and permissions.

Instead of: Kevin, Computer01, HR Department....AD sees: User Object, Computer Object, and Group Object. Very important for automation, permissions, and administration.

Taking User Object as an example. It has attributes. Username, Full Name, Password, Email Address, Department, Job Title, Phone Number, and so on....varies by place.

Attacks can target user accounts. Password spraying is when an attacker tries a small number of common passwords against many user accounts. Brute force attacks, phishing, etc are all common attacks.

A computer object is an AD object which represents a computer that has joined the domain. When a Windows PC named PC-01 joins a company domain: AD creates a computer object for it. They are managed through AD because they can centrally control, secure, and monitor thousands of devices from a single system.

Groups + Enterprise Permissions

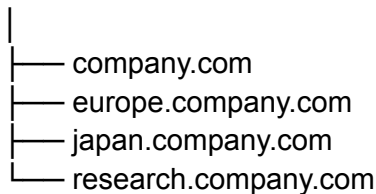
AD groups are collections of users, computers, or other groups that are managed together. Permissions are assigned to groups rather than individual users. This is known as Role-Based Access Control (RBAC).

Domain Users: Every normal user account is automatically a member of the Domain Users group. "Default employee"

Domain Admins: Administrators of a specific Active Directory domain. Think: company.com. A domain admin controls almost everything inside that domain.

Enterprise Admins: Administrators of the entire Active Directory forest.

Forest

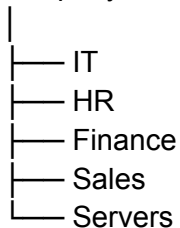


Each one of these domains has its own Domain Admins. But Enterprise Admins control ALL of them.

Organizational Units (OUs)

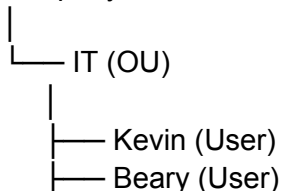
A container in AD used to organize users, computers, groups, and other objects into logical sections. Think: folder in AD

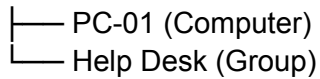
company.com



Each of these folders is an OU. In an OU: user objects, computer objects, groups, nested OUs..

company.com

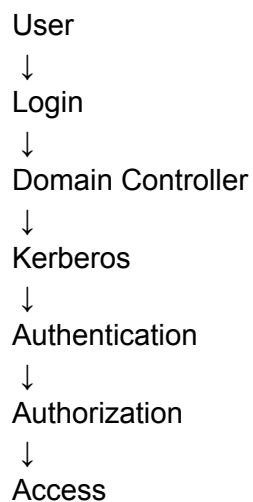




Group policies are applied to OUs by linking a Group Policy Object to an OU. Once linked, the policy automatically applies to the users and computers contained within the OU.

Identity + Authentication Flow

When a domain user logs in, AD performs authentication, authorization, and policy application before giving the user access.



LDAP (Lightweight Directory Access Protocol): Protocol used to communication with and query AD.

AD = directory/database

LDAP = language used to ask the directory questions